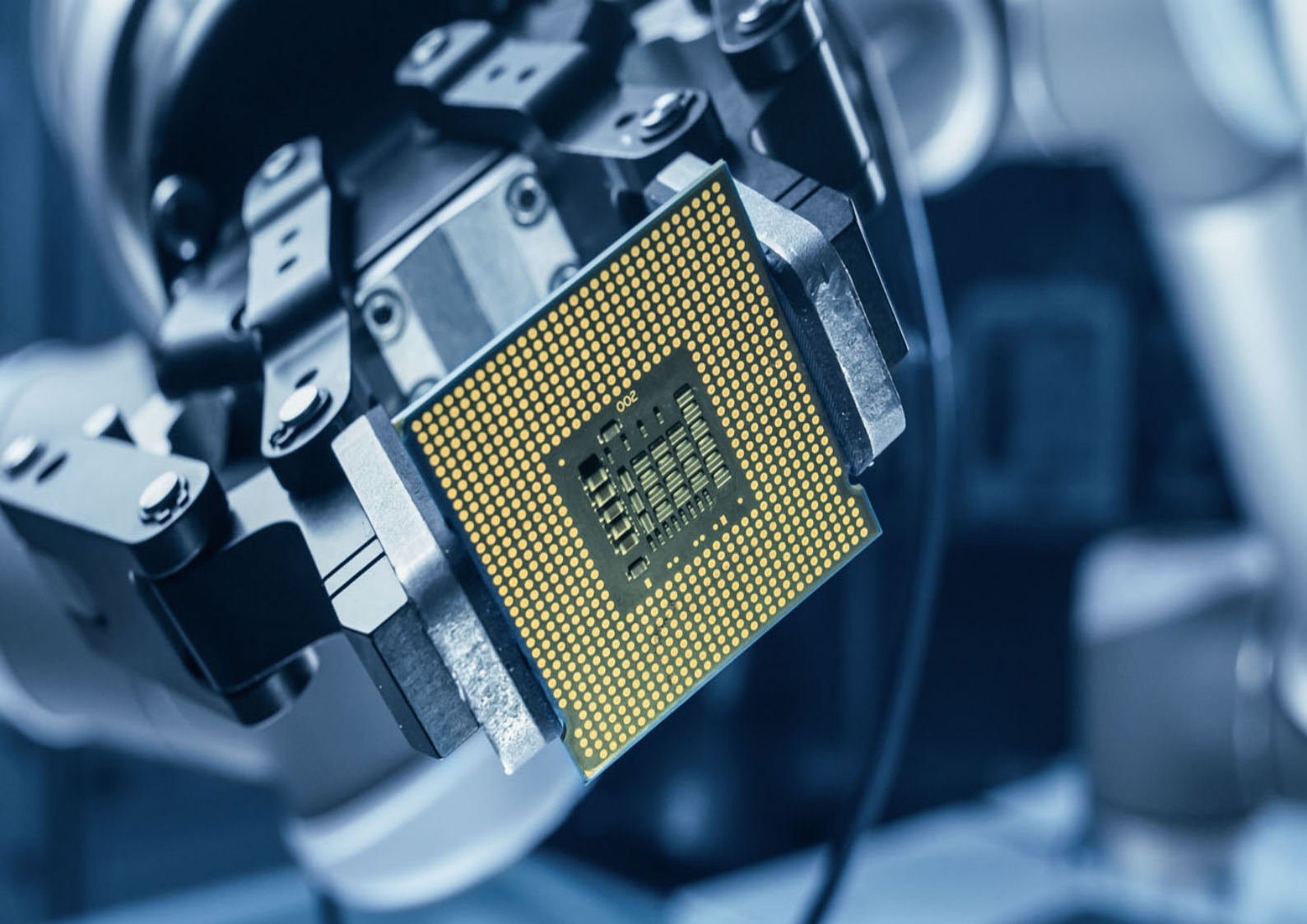




The SPICE Diode Model

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Introduction

This chapter presents a description of the SPICE diode model and methods for the extraction of its parameters. A comprehensive examination of this model will be given along with comparisons of the characteristics of a real diode and those produced by the model. These comparisons will be used to illustrate the model's accuracy and limitations. Based on the nature of the model equations, mathematical methods for the extraction of the SPICE parameters (with the exception of the noise parameters) will be presented. These methods are easily implemented for a Motorola MURH840CT rectifier as example. Throughout this chapter, references are made to the characteristics and behavior of an ideal diode model. The development of this model may be found in several of the many well-known texts on the subjects of semiconductor device theory and *pn* junction diodes.

The SPICE model of a *pn* junction diode consists of mathematical equations, parameters, and variables, all of which are designed to work together to simulate as accurately as possible the electrical characteristics of a real device. The equations and variables used for the model in the SPICE program are fixed and not easily modified. Therefore, the accuracy resulting from a SPICE simulation of a diode depends on the precise extraction methods which can be applied to this model, some of which yield more accurate results than others. The inherent success of a particular method depends for the most part on how well device physics and theory are utilized in the design of the model, and upon the availability of data taken from a real device.

The SPICE Diode Model

The parameters for the SPICE model of the diode are given in Table 3.1. The equations that use these parameters are divided into four model groups which are responsible for simulating various diode characteristics. These groups consist of the large-signal dc model, the small signal ac model, temperature and area effects, and the noise model.

The Large-signal DC Model

The large-signal behavior of the SPICE diode is characterized by the relationship between the dc current and voltage at its terminals. The parameters used to model this behavior are IS, RS, N, BV, and IBV. The parameter IS is the same as the reverse saturation current I_s for an ideal diode. The ohmic resistance RS is used to model the resistance of the metal contacts and the neutral regions under high-level injection. The emission coefficient N is used to modify the slope of the current versus voltage (I-V) characteristic curve. Finally, the parameters BV and IBV model the reverse breakdown behavior. Figure 3.1 shows the equivalent circuits of the SPICE diode for large-signal dc analysis. This circuit contains an internal diode D_1 , a series resistance having a value of RS, and a shunt conductance GMIN. The SPICE program adds this conductance, which is transparent to the user, around every internal *pn* junction to aid convergence. The program default value for GMIN is 10^{-12} mhos but can be set to any other nonzero value with a .OPTIONS line in the circuit file.

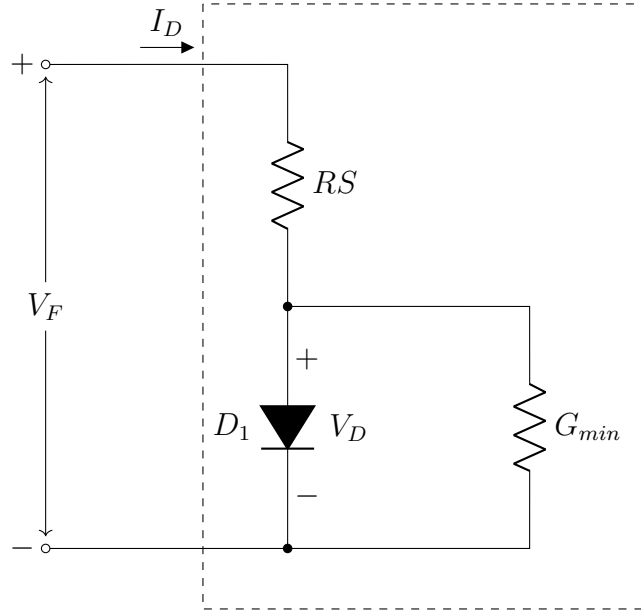


Figura 1: Amplificador multietapa

The dc model variables consists of the voltage across the external diode terminals V_F , the voltage across the interbal diode terminals V_D , and terminal current I_D . With this parameters and variables, the large-signal dc characteristics are modeled by the following equations.

$$V_F = R_S \cdot I_D + V_D \quad (1)$$

y

$$I_D = f(V_D) \quad (2)$$

Where four regions of operation describe the functional relationship between the internal diode voltage and the diode current.

$$I_{C1} = \frac{\beta}{\beta + 1} I_{ref} \quad (3)$$

Donde

$$I_{ref} = \frac{V_{CC} - V_{BE}}{R_1} \quad (4)$$

Para $\beta \gg 1$

$$I_{ref} \cong I_{C1} \quad (5)$$

Ejemplo

Diseñe una fuente de corriente Widlar para generar una corriente constante $I_o = 10 \mu\text{A}$. Asuma que $V_{CC} = 10 \text{ V}$, $V_{BE} = 0.7 \text{ V}$, $\beta = 125$. Use $V_T = 25 \text{ mV}$.

Para la fuente de corriente Widlar primero debemos decidir un valor apropiado para I_{ref} . Si escogemos $I_{ref} = 1 \text{ mA}$, entonces R_1 está dado por

$$R_1 = \frac{10 \text{ V} - 0.7 \text{ V}}{1 \text{ mA}} = 9.3 \text{ k}\Omega \quad (6)$$

El valor de R_E está dado por la ecuación (7)

$$R_E = \frac{25 \text{ mV}}{\left(1 + \frac{1}{125}\right) 10 \mu\text{A}} \ln \left(\frac{1 \text{ mA}}{10 \mu\text{A}} \right) = 11.5 \text{ k}\Omega \quad (7)$$

Es claro ver que el circuito de Widlar permite la generación de constantes pequeñas usando resistores relativamente pequeños.